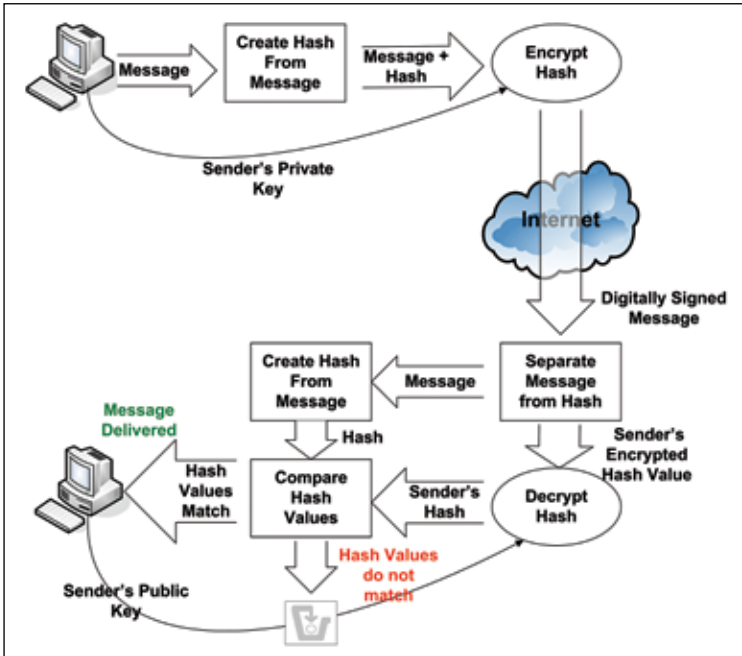




Hashing in action

The whole message including the digest is then encrypted using a one-time symmetric-key which is known only to Alex and Neil. Neil decrypts the message using the symmetric-key. He then decrypts the message digest using the public-key. He would then hash the original message using the same hash algorithm (whose name was appended in the message) with which it was previously hashed. If the evaluated digest and decrypted digest match, then the signature has been verified and the recipient would be sure that the message integrity has been preserved.

Another aspect of this system is the non-repudiation of digital signatures. Since the private-key is only privy to the sender, he can't deny signing the message. Also, a digital signature can be verified by anyone using the sender's public-key which is usually included in the digital signature format.

Electronic money

Electronic information has replaced cash for financial transactions between individuals for quite a long time now. Such a system uses cryptography to